

07 The Case Against Generation

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When a deterministic pixel predictor meets an open future, its optimum can be a picture of something that cannot happen. What sampling fixes, what embeddings avoid, and what each still owes.

Point a camera at a road and ask what the next second looks like.

A leaf on a tree at the edge of the frame will move. Which way it moves is unknowable, and it is also the least interesting fact available. Nothing you might decide to do depends on it.

But a model asked to predict the picture has to predict that leaf, because the leaf is made of pixels and pixels are what it is being marked on. Multiply that by every leaf, every ripple, every flicker of light on wet tarmac. Most of the capacity in the system has now gone to things that are hard to predict and do not matter.

That is the smaller of the two problems.

A deterministic average of two futures is neither

Here is the larger one.

A car is coming to a junction. It will turn left or it will turn right. You genuinely do not know which, and no amount of looking harder at the current frame will tell you.

Ask a deterministic model for one next picture and score it with squared pixel error, and there is a mathematically best answer. It is not either turn. It is the conditional average of them.

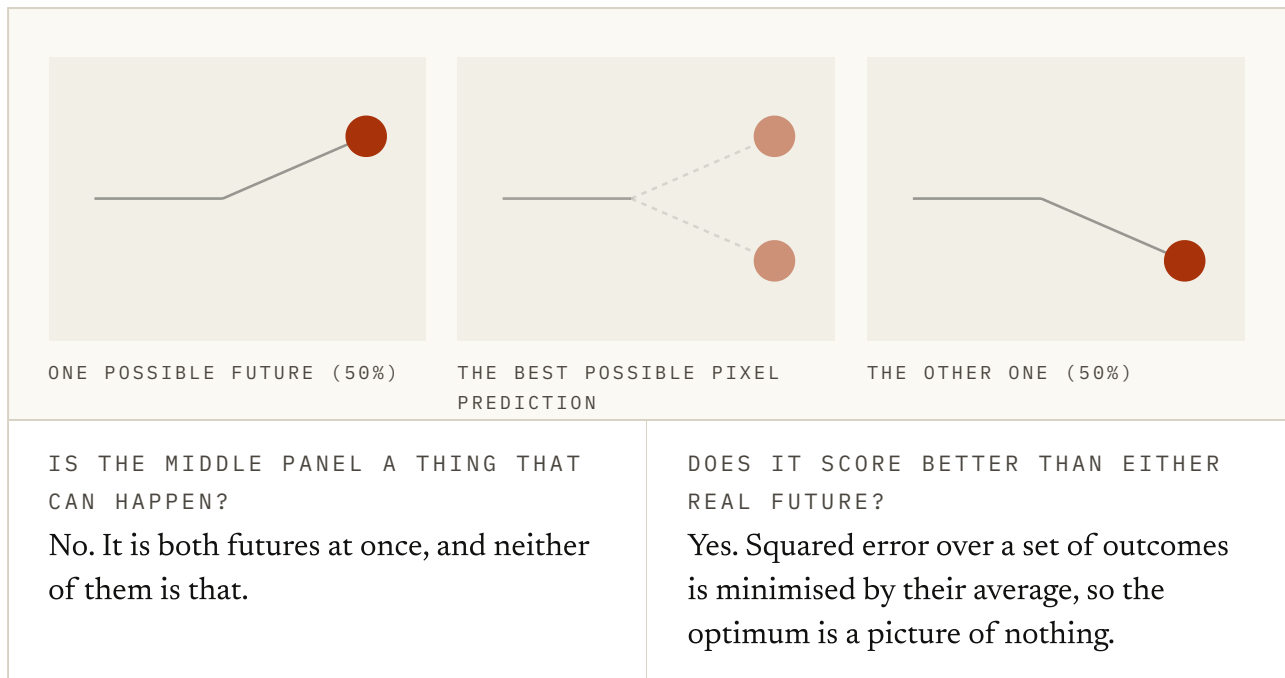


FIG. 7.1 The middle panel is not an impression of blurriness. It is the two futures drawn at their probabilities, which is their average, which is exactly what squared error is minimised by. Slide the chance towards either end and the optimum resolves; leave it near the middle and the best available prediction is a picture of something that cannot happen.

Read that middle panel again. It scores better than either of the real futures. A model that confidently predicted the left turn would be punished half the time; the ghost is punished a little all the time, and a little all the time wins.

That particular training signal is not asking for a plausible future. It is asking for the mean, and when the future is genuinely open, the mean can be a smear that never occurs.

A deterministic predictor trained with a mean-seeking pixel loss gets vague exactly when something interesting is about to happen. Modern generators often use stochastic objectives precisely to avoid that failure.

Sampling is a real counterargument

Pixel prediction does not have to emit one answer. Stochastic video models such as SV2P and Denton and Fergus's model learn a distribution over futures, then draw from it. Each draw can be sharp and plausible even when the model is unsure which turn will occur.

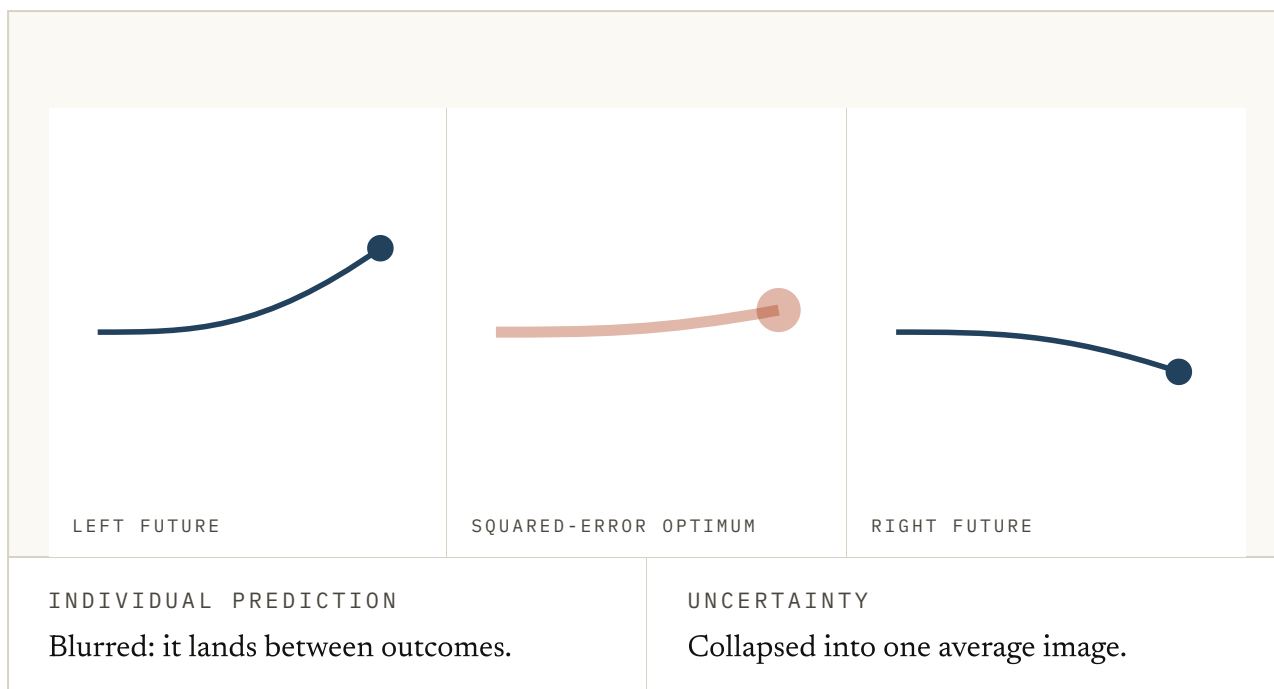


FIG. 7.2 Switch from the conditional mean to samples from the same two-outcome distribution. Sampling removes the impossible in-between frame and preserves uncertainty across draws. It does not reveal which draw the world will choose, and a planner still has to value risk across the whole distribution.

This kills the blur objection. It does not kill the rest of the case against generation. Sampling fixes the form of the output; it does not make irrelevant texture cheap, make rare futures well estimated, or tell a decision maker which possible future matters. Embedding prediction changes the target. Stochastic generation keeps the target and represents its uncertainty better.

Predict the description instead

The proposal is short. Stop predicting the picture.

Encode what you can see into a compact description (an *embedding*: a short list of numbers standing in for something bigger, arranged so that similar things land near each other). Encode the part you are predicting into one too. Then train the model to predict the second description from the first, and never decode anything back into pixels at all.

This buys two things at once.

The leaf problem goes away, because the description was never obliged to record which way the leaf went. If a detail does not survive the encoding, nothing downstream is graded on it.

The averaging problem can get smaller, because you are no longer averaging pictures. The descriptions of a car turning left and right can share most of their coordinates, leaving the unresolved part explicit or irrelevant. A feature space can still be multimodal, however; changing the representation does not abolish uncertainty.

This is the JEPA family, and the name says the architecture: joint embedding, because both sides get embedded, and predictive, because the training signal is one embedding predicting another.

The bill for that

Now the part that is usually skipped, and it is not a small part.

Predicting pixels has one enormous virtue. The target is fixed. The next frame is the next frame, and the model cannot make it easier.

Predict an embedding and that stops being true, because the thing you are predicting is produced by a network that is also being trained. The network can get a perfect score while learning nothing. It makes every description identical. The prediction is then always exactly right, the loss goes to zero, and the representation says nothing about anything.

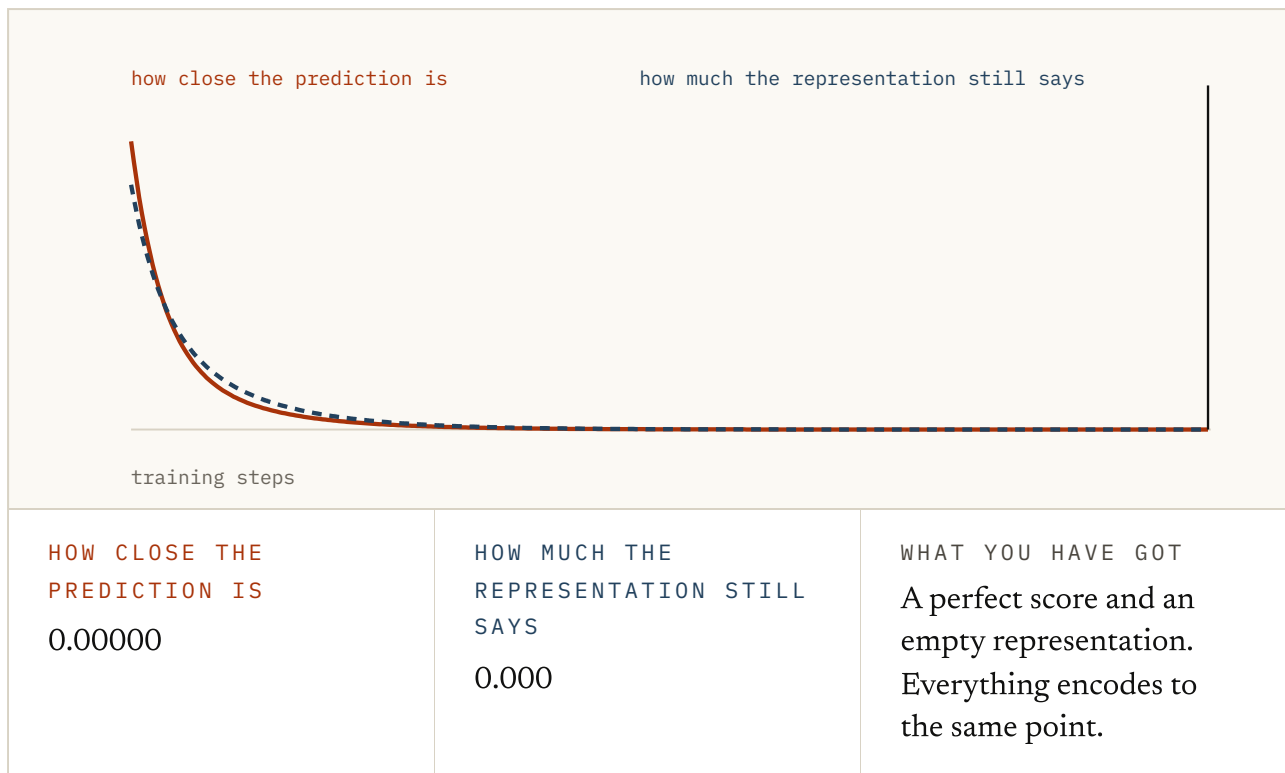


FIG. 7.3 A two-by-two encoder trained on four pairs by gradient descent. The numbers come straight from the arithmetic. Without the safeguard, the loss and the representation both reach zero. Turn the safeguard on and the loss never gets close to zero. The useful run is the one that looks worse.

Everything difficult about these methods lives in preventing that. Keep a second copy of the encoder that updates slowly (usually called a *target encoder*: it trails the one being trained, so the thing being predicted cannot sprint away from the thing predicting it), so the target cannot run away. Push different inputs apart explicitly. Penalise descriptions whose components all say the same thing. There are several families of answer and none of them is elegant.

Notice what has happened to the loss. It is no longer a measurement you can read. A pixel loss of zero means the model predicted the frame. An embedding loss of zero might mean it understands everything, or nothing at all, and the number will not tell

you which. The score stopped being trustworthy at the same moment it stopped being about pixels.

What this does not settle

It would be tidy to end with generation being a mistake. It is not.

Systems that predict pixels are the ones producing worlds you can walk around in, and they work. Whatever is being wasted on leaves, they are delivering something that no amount of embedding prediction delivers, which is an actual picture. If you want to see the future rather than reason about it, somebody has to draw it.

So the honest statement of the argument is narrower than the slogan. Predicting appearances spends most of its capacity on things no decision depends on. And when the future is uncertain, it is trained towards an image of something that will not happen. If what you want is a compact state to act on, that is a bad way to get one. If what you want is a picture, it is the only way anyone has.

The two goals were never the same goal. Most of the confusion in this corner of the field is people comparing systems that were built for different ones.

Try it

1 A deterministic predictor uses squared pixel error and a car turns left or right with equal probability. What is its optimum?

- a. The average of the two, which is a picture of neither
- b. Whichever was more common in training
- c. The left turn
- d. The right turn

ANSWER a. The average of the two, which is a picture of neither. Squared error over a set of outcomes is minimised by their mean. Committing to one turn is punished half the time; the smear is punished a little all the time, and that wins.

2 A stochastic video model now draws sharp left and right turns instead of their blur. What has it fixed, and what has it not?

- a. It removed the need for a pixel target
- b. It proved embedding prediction is unnecessary
- c. It has learned which turn the real car will choose
- d. It fixed the impossible average but still has to represent probabilities and decision risk

ANSWER d. It fixed the impossible average but still has to represent probabilities and decision risk. Sampling is a real solution to blur: each draw can be plausible. It does not identify which draw reality will select or tell a planner how to value the distribution.

3 A leaf at the edge of the frame moves unpredictably. Why does a pixel-predicting model spend capacity on it?

- a. Leaves are important for scene understanding
- b. The leaf is made of pixels, and pixels are what the model is marked on
- c. It cannot tell leaves from cars
- d. The encoder forces it to

ANSWER b. The leaf is made of pixels, and pixels are what the model is marked on. Nothing tells the objective which pixels matter. Hard-to-predict and irrelevant is the worst combination, and it is most of the frame.

4 What does predicting an embedding rather than a picture do to the leaf problem?

- a. Nothing, the leaf is still in the input
- b. It makes the leaf easier to predict
- c. If a detail does not survive the encoding, nothing downstream is graded on it
- d. It moves the problem to the decoder

ANSWER c. If a detail does not survive the encoding, nothing downstream is graded on it. The description was never obliged to record which way the leaf went, so the model is not punished for failing to say.

5 Predicting pixels has one large virtue that predicting embeddings gives up. What is it?

- a. The target is fixed: the model cannot make the next frame easier
- b. It needs less data
- c. It generalises better
- d. It is faster to compute

ANSWER a. The target is fixed: the model cannot make the next frame easier. The moment the target is produced by a network that is also being trained, the target can move, and there is a way to make it move somewhere very convenient.

6 What is collapse?

- a. The model forgetting earlier training
- b. Gradients vanishing in deep layers
- c. The loss diverging to infinity
- d. Every input encoding to the same description, so the prediction is always right and the representation says nothing

ANSWER d. Every input encoding to the same description, so the prediction is always right and the representation says nothing. It is a perfect score obtained by learning nothing, and it is the cheapest solution available unless something is specifically stopping it.

7 In Figure 7.3, the run with the safeguard has a worse loss. What does that tell you?

- a. The safeguard is badly tuned
- b. An embedding loss is no longer a number you can read off as quality
- c. The model needs more training
- d. The safeguard should be removed once training stabilises

ANSWER b. An embedding loss is no longer a number you can read off as quality. A pixel loss of zero means the frame was predicted. An embedding loss of zero might mean everything or nothing, and the number cannot tell you which.

8 Does this argument show that generating pixels is a mistake?

- a. No, because collapse makes embeddings unusable
- b. Yes, it is strictly worse
- c. No. It is a bad way to get a compact state to act on, and the only way anyone has to get an actual picture
- d. Yes, except for very short videos

ANSWER c. No. It is a bad way to get a compact state to act on, and the only way anyone has to get an actual picture. The two goals were never the same goal, and most of the confusion here is people comparing systems built for different ones.

FIG. 7.4 Eight questions on the argument and its bill. The last two are the ones that stop this reading as a knock-down case.

Which leaves the systems that went the other way, and went hard.

Thanks for reading. Chapter 8 is about what happens when you scale pixel prediction until the result is something you can steer with a keyboard. Also about what "it learned physics" does and does not mean when someone says it about a video model.

Sources

Stochastic Variational Video Prediction (SV2P) BABAEIZADEH ET AL., 2017 ↗

The real counterargument to deterministic blur: learn a distribution and draw distinct plausible futures instead of returning their pixelwise mean.

<https://arxiv.org/abs/1710.11252>

A Path Towards Autonomous Machine Intelligence LECUN, 2022 ↗

The position paper the whole argument comes from, including why predicting appearances is the wrong job for a system meant to act.

<https://openreview.net/forum?id=BZ5a1r-kVsf>

I-JEPA ASSRAN ET AL., 2023 ↗

Hide part of an image and predict the embedding of the missing piece rather than redrawing it. The clean statement of the method.

<https://arxiv.org/abs/2301.08243>

V-JEPA 2 META AI, 2025 ↗

The video version, pre-trained without actions and then post-trained for control, which is where the argument meets a robot.

<https://ai.meta.com/blog/v-jepa-2-world-model-benchmarks/>

Bootstrap Your Own Latent (BYOL) GRILL ET AL., 2020 ↗

The slowly-updating target copy, and the result that made people believe you could avoid collapse without pushing things apart.

<https://arxiv.org/abs/2006.07733>

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The explicit approach: penalise a representation whose components have collapsed or duplicated each other. Figure 7.3's safeguard, done properly.

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A Cookbook of Self-Supervised Learning BALESTRIERO ET AL., 2023 ↗

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<https://arxiv.org/abs/2304.12210>

Masked Autoencoders Are Scalable Vision Learners HE ET AL., 2021 ↗

The counter-example worth holding on to: reconstruct the pixels of the masked part, and it works very well anyway.

<https://arxiv.org/abs/2111.06377>

Learning and Leveraging World Models in Visual Representation Learning

GARRIDO ET AL., 2024 ↗

What the embedding-prediction objective turns out to have learned, tested rather than asserted.

<https://arxiv.org/abs/2404.08471>

FIG. 7.5 Ordered for someone building on this rather than for historical completeness.